

DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING
INT-375

PROJECT REPORT

(Project Semester January-April 2025)

**COVID-19_Outcomes_by_Vaccination_Status_-_Historical Data
Analysis**

Submitted by

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Programme and Section- BTECH(CSE) & K23GD

Course Code -INT375

Under the Guidance of

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CERTIFICATE

This is to certify that **Bablu Kumar Sharma** bearing Registration no. **12313003** has completed **INT-375** project titled, “COVID-19_Outcomes_by_Vaccination_Status_-_Historical Analysis” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Mrs. Baljinder Kaur

School of Computer Science and Engineering
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Date: 12-04-25

DECLARATION

I **Bablu Kumar Sharma**, student of **B.Tech CSE - K23GD** under CSE Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12-04-25

Signature

Registration No.- **12313003**

Name- **Bablu Kumar Sharma**

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty guide **Mrs. Baljinder Kaur** for her valuable guidance, support, and encouragement throughout this project. I am thankful to **Lovely Professional University** for providing the necessary resources and environment for conducting this project successfully. I also extend my appreciation to all those who directly or indirectly contributed to the completion of this report.

TABLE OF CONTENTS

1. Introduction
2. Source of dataset
3. EDA process
4. Analysis on dataset (for each analysis)
 - i. COVID-19 Trends Over Time by Vaccination Status
 - ii. Effectiveness of Vaccination in Reducing Severe Outcomes
 - iii. Risk Reduction Analysis
 - iv. Age Group Vulnerability Analysis
 - v. Temporal Trends Analysis
 - vi. Age Group Vulnerability Analysis
 - vii. Numerical Feature Trend Analysis
 - viii. Outcome Rate Distribution Analysis Using Box Plots
5. Conclusion
6. Future scope
7. References.

1. Introduction

In the wake of the global COVID-19 pandemic, data-driven insights have become crucial for public health strategies and vaccine policy decisions. This project presents a comprehensive analysis of COVID-19 outcomes by vaccination status in the United States, using a cleaned and structured dataset sourced from public health records. The analysis was performed using Python and data visualization libraries such as Matplotlib and Seaborn.

The primary objective is to explore the relationship between vaccination status and key COVID-19 outcomes such as cases, hospitalizations, and deaths. By examining trends across different age groups and time periods, the report aims to uncover patterns that inform the effectiveness of vaccination campaigns. The cleaned dataset, visualized through graphs and summary tables, provides a valuable tool for health analysts, researchers, and policy-makers to evaluate public health responses and plan for future contingencies.

2. Source of Dataset

The dataset used in this project comes from public health records and is titled COVID-19 Outcomes by Vaccination Status - Historical. It includes information about COVID-19 cases, hospitalizations, and deaths, grouped by vaccination status such as Unvaccinated, Vaccinated, and Boosted. The data also includes different age groups and is updated weekly.

Since the dataset covers several months, it helps us see how outcomes have changed over time. This makes it useful for understanding how vaccines have helped reduce the impact of COVID-19. The dataset is a good source for analysing trends and making informed decisions in healthcare and public safety.

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3. EDA Process

In this project, EDA helped clean and organize the COVID-19 data so it could be used effectively for analysis and visualization. The following steps were carried out:

- Checked for missing values and filled or removed them where needed
- Standardized column names by converting them to lowercase and replacing spaces with underscores
- Converted date columns to proper datetime format for easier sorting and filtering

- Removed duplicate records to avoid errors in analysis
- Cleaned up incorrect or invalid entries in the age group and vaccination status columns
- Dropped columns with too many missing values that could not be used
- Sorted the data by reporting dates to observe changes over time
- Filtered the dataset to include only valid age groups for more accurate insight

Outcome:-

```

Column names and data types:
Outcome                object
Week End               object
Age Group              object
Unvaccinated Rate      float64
Vaccinated Rate        float64
Boosted Rate           float64
Crude Vaccinated Ratio float64
Crude Boosted Ratio    float64
Age-Adjusted Unvaccinated Rate float64
Age-Adjusted Vaccinated Rate float64
Age-Adjusted Boosted Rate float64
Age-Adjusted Vaccinated Ratio float64
Age-Adjusted Boosted Ratio float64
Population Unvaccinated int64
Population Vaccinated  float64
Population Boosted     float64
Outcome Unvaccinated   int64
Outcome Vaccinated     float64
Outcome Boosted        float64
Age Group Min          int64
Age Group Max          int64
dtype: object

Preview of dataset:
   Outcome  Week End Age Group Unvaccinated Rate Vaccinated Rate Boosted Rate ... Population Boosted Outcome Unvaccinated Outcome Vaccinated Outcome Boosted Age Group Min Age Group Max
0 Deaths  10/01/2022   All      0.3           0.1      0.9 ...      1037067.0           2           1.0           9.0      999      999
1 Deaths  02/04/2023  50-64      2.9           0.8      0.4 ...      246849.0           1           1.0           1.0       50       64
2 Deaths  09/10/2022   5-11      0.0           0.0      0.0 ...      17689.0           0           0.0           0.0        5       11
3 Deaths  11/20/2021   5-11      0.0           0.0      NaN ...           NaN           0           0.0           NaN        5       11
4 Deaths  04/09/2022  18-29      0.0           0.0      0.0 ...      147442.0           0           0.0           0.0       18       29

[5 rows x 21 columns]

Missing values per column:
Outcome                0
Week End               0
Age Group              0
Unvaccinated Rate      327
Vaccinated Rate        1224
Boosted Rate           1433
Crude Vaccinated Ratio 1433
Crude Boosted Ratio    1992

```

[5 rows x 21 columns]

Missing values per column:

Outcome	0
Week End	0
Age Group	0
Unvaccinated Rate	0
Vaccinated Rate	327
Boosted Rate	1224
Crude Vaccinated Ratio	1433
Crude Boosted Ratio	1992
Age-Adjusted Unvaccinated Rate	3336
Age-Adjusted Vaccinated Rate	3336
Age-Adjusted Boosted Rate	3423
Age-Adjusted Vaccinated Ratio	3373
Age-Adjusted Boosted Ratio	3441
Population Unvaccinated	0
Population Vaccinated	540
Population Boosted	1224
Outcome Unvaccinated	0
Outcome Vaccinated	327
Outcome Boosted	1224
Age Group Min	0
Age Group Max	0
dtype: int64	

Unique values per column:

Outcome	3
Week End	140
Age Group	9
Unvaccinated Rate	1287
Vaccinated Rate	915
Boosted Rate	834
Crude Vaccinated Ratio	241
Crude Boosted Ratio	163
Age-Adjusted Unvaccinated Rate	272
Age-Adjusted Vaccinated Rate	207
Age-Adjusted Boosted Rate	176
Age-Adjusted Vaccinated Ratio	109
Age-Adjusted Boosted Ratio	86
Population Unvaccinated	1088
Population Vaccinated	963
Population Boosted	785


```

Column names after cleaning:

['outcome', 'week_end', 'age_group', 'unvaccinated_rate', 'vaccinated_rate', 'boosted_rate', 'crude_vaccinated_ratio',
 'age-adjusted_boosted_rate', 'age-adjusted_vaccinated_ratio', 'age-adjusted_boosted_ratio', 'population_unvaccinated',
 'population_vaccinated', 'population_boosted', 'outcome_unvaccinated', 'outcome_vaccinated', 'outcome_boosted',
 'age_group_min', 'age_group_max']

Missing values before filling:

outcome                0
week_end               0
age_group              0
unvaccinated_rate      0
vaccinated_rate       327
boosted_rate          1224
crude_vaccinated_ratio 1433
crude_boosted_ratio    1992
age-adjusted_unvaccinated_rate 3336
age-adjusted_vaccinated_rate 3336
age-adjusted_boosted_rate 3423
age-adjusted_vaccinated_ratio 3373
age-adjusted_boosted_ratio 3441
population_unvaccinated 0
population_vaccinated   540
population_boosted     1224
outcome_unvaccinated    0
outcome_vaccinated      327
outcome_boosted        1224
age_group_min          0
age_group_max          0
dtype: int64
d:\4 semester\INT 375\project\Obj1.py:33: FutureWarning: Series.fillna with 'method' is deprecated and will raise in a future version
df['week_end'] = df['week_end'].fillna(method='ffill') # Forward fill missing dates

Unique values in 'age_group' column after cleaning:

['All' '50-64' '18-29' '30-49' '80+' '65-79']

Missing values after filling:

outcome                0
week_end               0
age_group              0
unvaccinated_rate      0
vaccinated_rate        0
boosted_rate           0

```

4. Analysis on dataset (for each analysis)

4.1 COVID-19 Trends Over Time by Vaccination Status

i. Introduction

This part of the analysis focuses on how COVID-19 trends—specifically cases, hospitalizations, and deaths—have changed over time based on vaccination status. It helps us understand the effectiveness of vaccines in reducing the spread and severity of the virus. By comparing these outcomes across unvaccinated, vaccinated, and boosted groups, we can observe the impact of vaccination efforts over the course of the pandemic.

ii. General Description

The dataset contains weekly data showing rates of COVID-19 cases, hospitalizations, and deaths per 100,000 people for three groups: Unvaccinated, Vaccinated, and Boosted. These outcomes are recorded over a period of time, allowing us to track and compare

trends. This time-based comparison provides insights into how different levels of vaccination protection affected health outcomes during different stages of the pandemic.

iii. Specific Requirements, Functions and Tools

The analysis was performed using Python, with the help of libraries like Pandas and Matplotlib. Key steps and techniques included:

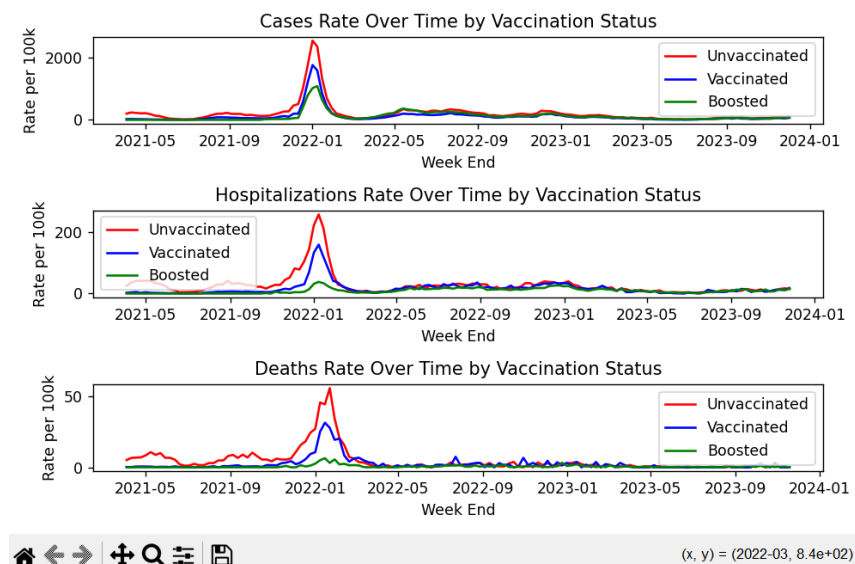
- **Filtering** the dataset to include only relevant outcomes (cases, hospitalizations, deaths)
- **Grouping** the data by week and outcome using `groupby()` to calculate average rates
- **Plotting** line graphs for each outcome over time for all vaccination statuses
- **Customizing** each subplot for clear visual comparison

iv. Analysis Results

The results of the analysis show a clear difference in outcomes between vaccination groups. Unvaccinated individuals consistently had higher rates of cases, hospitalizations, and deaths across most weeks. Vaccinated individuals had lower rates, and boosted individuals showed the lowest rates throughout the timeline.

This trend highlights the effectiveness of vaccines, especially booster shots, in reducing severe outcomes. It also supports public health efforts that encourage full vaccination and boosters as a key strategy in managing the impact of COVID-19

v. Visualization



4.2 Effectiveness of Vaccination in Reducing Severe Outcomes

i. Introduction

This analysis assesses how vaccination impacts the severity of COVID-19 outcomes, focusing on hospitalizations and deaths. The aim is to determine whether higher vaccination levels correlate with lower severe outcome rates.

ii. General Description

The dataset records weekly hospitalization and death rates by vaccination status—Unvaccinated, Vaccinated, and Boosted. This allows for a direct comparison of outcome severity across immunization levels.

iii. Specific Requirements, Functions and Formulas

Using Python (pandas, seaborn, matplotlib), we:

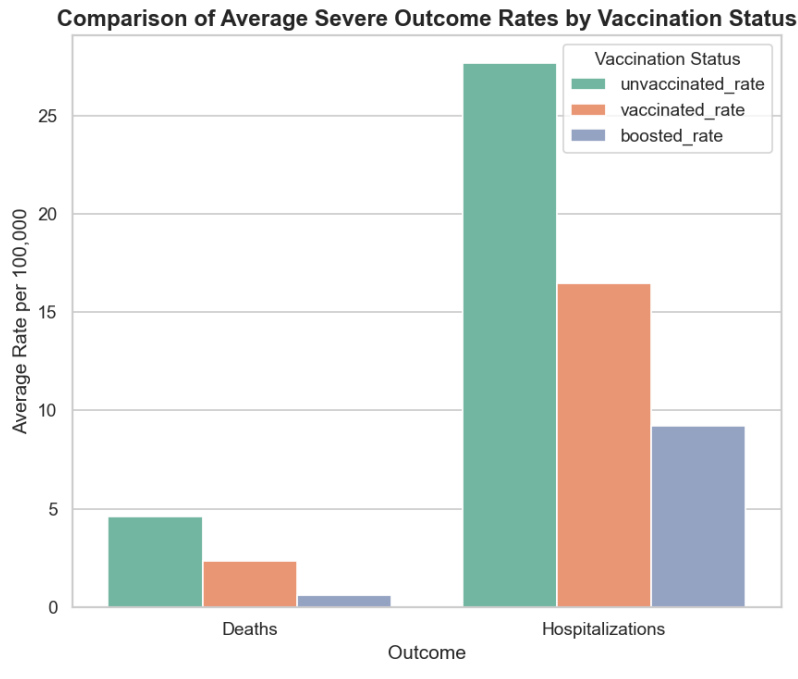
- Filtered for severe outcomes (Hospitalizations, Deaths)
- Calculated average rates per vaccination status
- Reshaped data with melt() for visualization
- Created a grouped bar chart using seaborn bar plot()

These steps quantified and visually compared severity based on vaccination coverage.

iv. Analysis Results

Findings show a clear decline in severe outcome rates with increased vaccination. Unvaccinated individuals had the highest rates, while boosted individuals had the lowest. This pattern confirms the effectiveness of vaccines in reducing critical health risks.

.v. Visualization



4.3 Risk Reduction Analysis

i. Introduction

This analysis quantifies how vaccination reduces the risk of adverse health outcomes due to COVID-19. By calculating outcome rates per 100,000 population, the objective is to assess the relative risk for unvaccinated, vaccinated, and boosted individuals.

ii. General Description

The dataset includes weekly counts of outcomes and the respective population sizes for each vaccination group. Rates per 100,000 were computed to standardize comparisons across varying population sizes and identify patterns in outcome severity linked to vaccination status.

iii. Specific Requirements, Functions and Formulas

Using Python (`pandas`, `matplotlib`), the following steps were performed:

- **Rate Calculation:** Outcome rates per 100,000 were computed for each vaccination group.
- **Aggregation:** Average rates by outcome type were derived using `groupby()` and `mean()`.
- **Visualization:** A multi-line chart was created to compare average rates across outcomes and vaccination statuses.

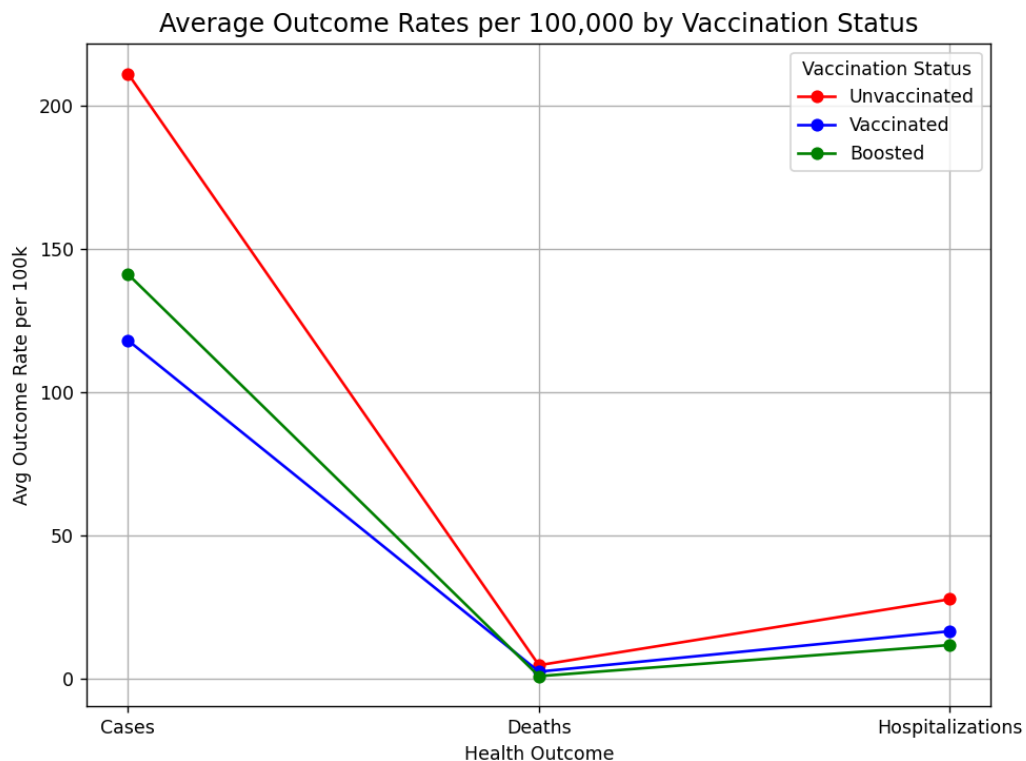
Key functions used:

- Mathematical operations for rate calculation
- `groupby()` for structured aggregation
- `matplotlib.pyplot` for custom line plotting

iv. Analysis Results

The analysis reveals that outcome rates consistently decline with increasing vaccination coverage. Unvaccinated individuals show the highest average risk, followed by vaccinated individuals, with boosted individuals having the lowest risk. This trend reinforces the role of vaccines in risk reduction.

v. Visualization



4.4 Age Group Vulnerability Analysis

i. Introduction

This analysis identifies which age groups are most affected by COVID-19 in terms of cases, hospitalizations, and deaths.

ii. General Description

The dataset contains outcome data segmented by age group and vaccination status. Total counts were calculated to assess vulnerability across demographics.

iii. Specific Requirements, Functions and Formulas

Using Python:

- Filtered data by outcome type
- Aggregated totals per age group
- Created pie charts to show proportional impact

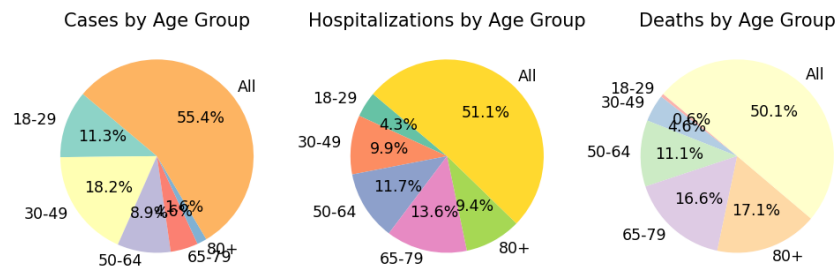
Key functions: `groupby()`, `sum()`, and `matplotlib.pyplot.pie()`.

iv. Analysis Results

Results show younger groups report more cases, while older groups have higher shares of hospitalizations and deaths, confirming increased risk with age.

v. Visualization

Distribution of Outcomes by Age Group



4.5 Temporal Trends Analysis

i. Introduction

This analysis tracks weekly changes in COVID-19 cases to identify trends such as outbreak peaks, seasonal patterns, and the potential impact of vaccination or interventions.

ii. General Description

The dataset includes weekly case counts by vaccination status. By aggregating data over time, the analysis highlights how infection rates evolve across groups.

iii. Specific Requirements, Functions and Formulas

Using Python:

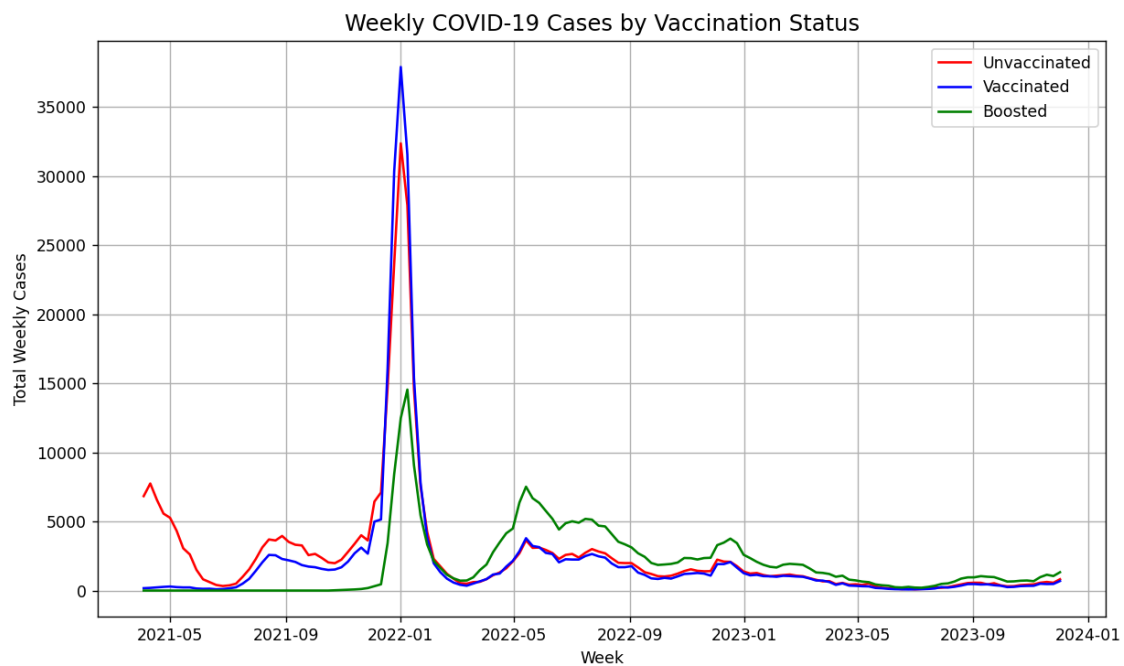
- Converted `week_end` to `datetime` for proper time-series analysis
- Grouped case data weekly
- Used `matplotlib.pyplot.plot()` to visualize trends across vaccination categories

Key functions: `groupby()`, `sum()`, and `plot()`.

iv. Analysis Results

The visualization reveals clear case surges during specific periods, with unvaccinated individuals consistently showing higher case counts. Vaccinated and boosted groups experience comparatively lower peaks, reflecting vaccine impact.

v. Visualization



4.6 Correlation Analysis of Deaths by Vaccination Status

i. Introduction

This analysis visualizes the correlation between vaccination status and COVID-19-related deaths to assess the relative burden among different population groups.

ii. General Description

The dataset includes cumulative death counts across unvaccinated, vaccinated, and boosted individuals. The aim is to show the proportional impact of each group on overall mortality.

iii. Specific Requirements, Functions and Formulas

Using Python:

- Filtered for the “Deaths” outcome
- Aggregated total deaths by vaccination status
- Used a donut chart (a stylized pie chart) for visual clarity

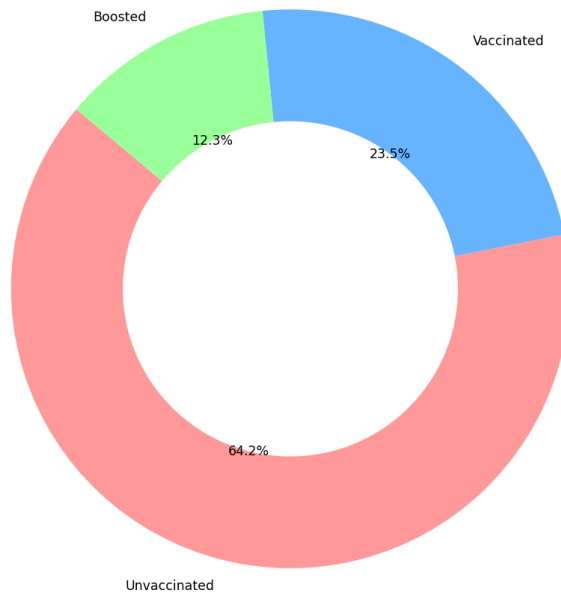
Key methods: `sum()`, `matplotlib.pyplot.pie()` with `wedgeprops={'width': 0.4}` for donut styling.

iv. Analysis Results

The visualization shows that unvaccinated individuals account for the highest proportion of deaths. This underlines the critical role of vaccination and booster doses in reducing fatality risk.

v. Visualization

Proportion of Deaths by Vaccination Status



4.7 Numerical Feature Trend Analysis

i. Introduction

This analysis visualizes the trends of key numerical metrics related to COVID-19 outcomes and vaccination statistics across records, enabling pattern recognition and outlier detection.

ii. General Description

The dataset includes various numerical attributes such as outcome rates, population counts, and crude vaccination ratios. By tracking these values across records, the analysis uncovers underlying fluctuations and correlations.

iii. Specific Requirements, Functions and Formulas

Using Python:

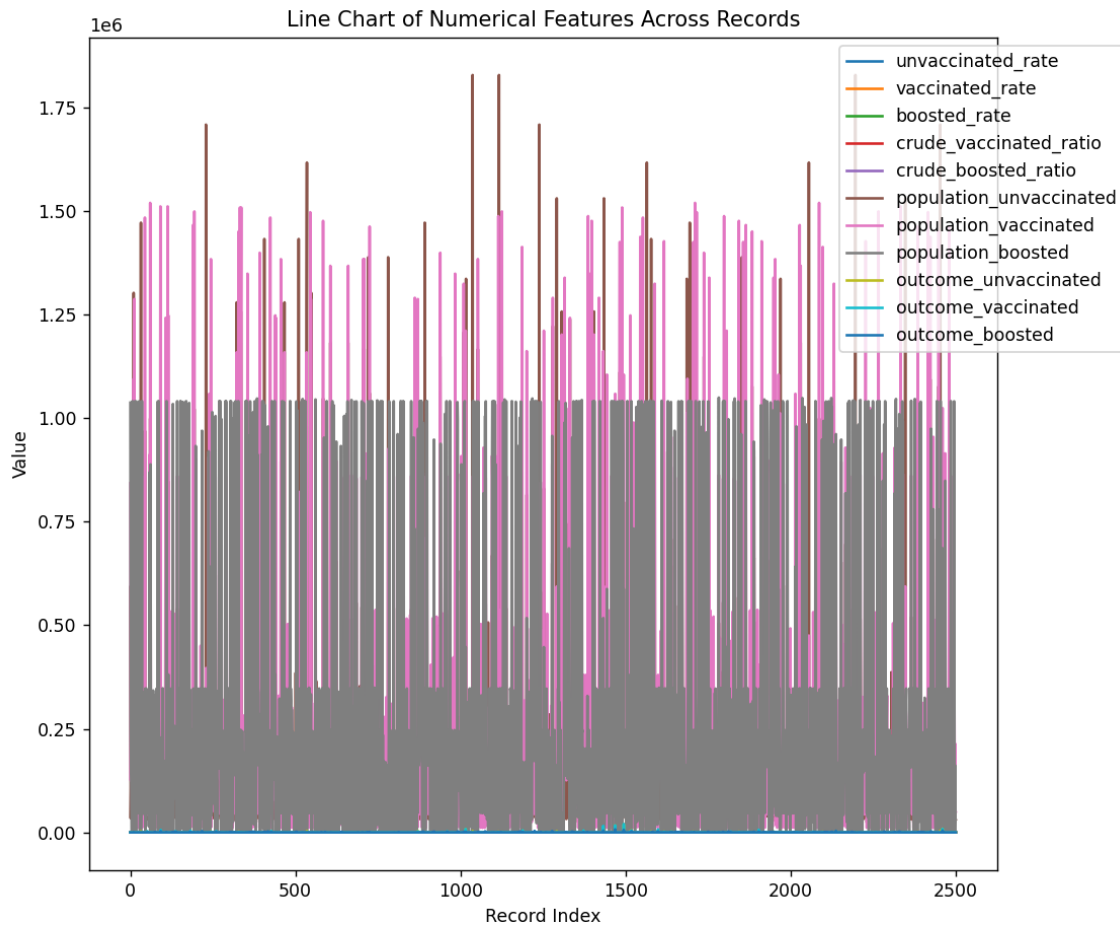
- Selected relevant numerical columns
- Reshaped the data using `melt()` for long-form plotting
- Utilized a multi-line chart for side-by-side trend visualization

Key functions: `melt()`, `plot()` from `matplotlib`.

iv. Analysis Results

The line chart reveals distinct trends and relative movement of features such as infection rates and vaccination coverage. Some variables remain stable, while others fluctuate significantly, indicating areas needing further investigation.

v. Visualization



4.8 Outcome Rate Distribution Analysis Using Box Plots

i. Introduction

This analysis investigates the distribution of COVID-19 outcome rates across different vaccination statuses (unvaccinated, vaccinated, boosted) using statistical visualization. Box plots are employed to highlight central tendencies, variability, and outliers in the outcome rates across groups.

ii. General Description

The dataset contains COVID-19 outcome counts (cases, hospitalizations, deaths) alongside population sizes for each vaccination status. To fairly compare across groups, outcome rates are normalized per 100,000 individuals. The analysis reshapes the data to facilitate grouped box plot visualization and applies a log scale to accommodate varying rate magnitudes.

iii. Specific Requirements, Functions and Formulas

Using Python:

- Calculated normalized outcome rates using the formula:

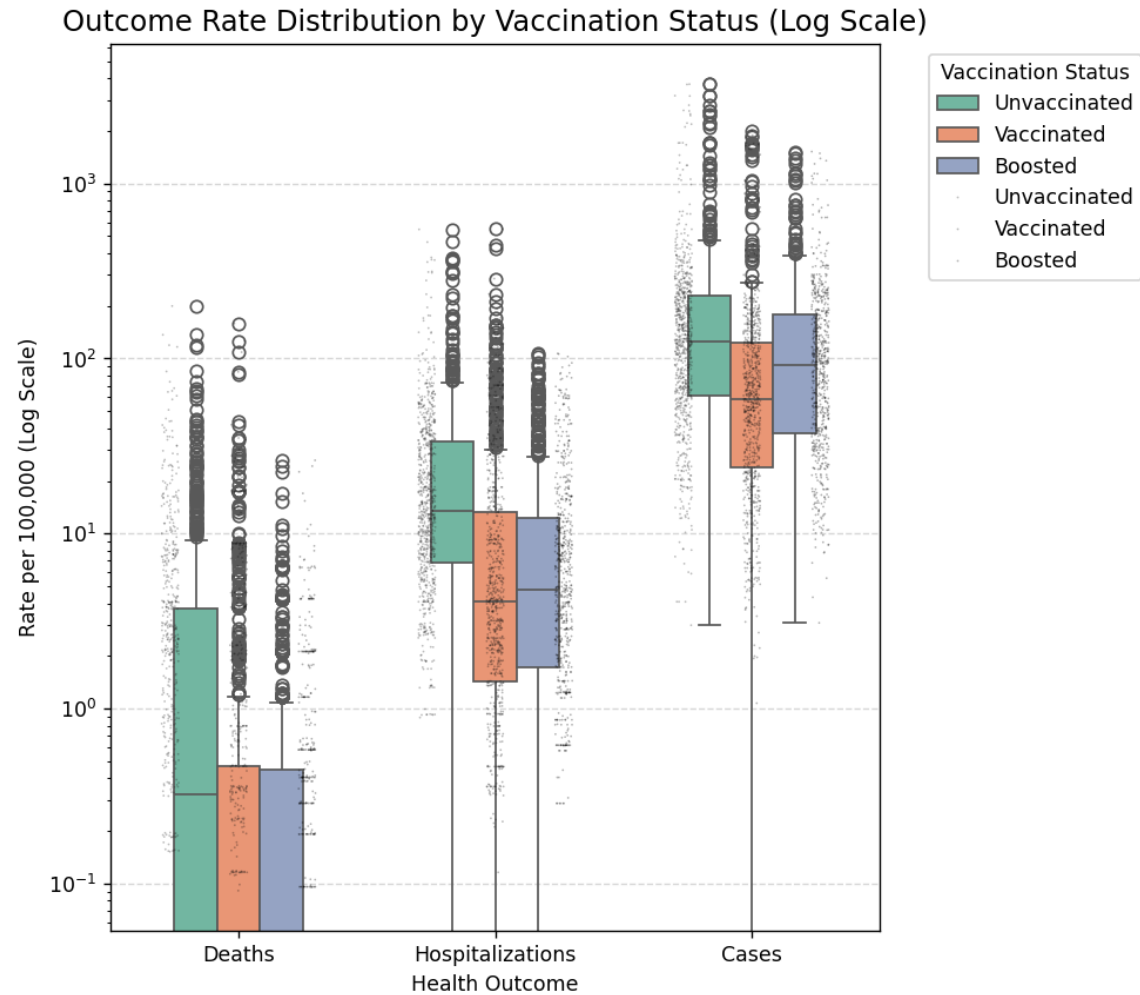
```
rate = (outcome_count / population) * 100,000
```
- Used `melt()` to reshape data from wide to long format for plotting
- Visualized with `seaborn.boxplot()` and enhanced distribution clarity using `seaborn.stripplot()`
- Applied logarithmic scaling with `plt.yscale("log")` to manage skewed distributions

Key functions: `melt()`, `boxplot()`, `stripplot()`, `yscale("log")`

iv. Analysis Results

The resulting visualization reveals clear differences in outcome rate distributions among the three vaccination groups. Unvaccinated individuals exhibit significantly higher median and outlier rates across all outcomes. The log scale effectively captures both subtle variations and extreme values. Booster recipients show the most consistent and lowest risk levels, while unvaccinated groups display greater variability and frequent outliers —signaling elevated vulnerability and inconsistency.

v. Visualization



5. Conclusion:

The exploratory data analysis of COVID-19 outcomes by vaccination status offers compelling insights into the trends, disparities, and effectiveness of vaccination efforts. Through a series of structured visualizations and statistical examinations, the following key conclusions emerged:

- **Vaccination Impact:** Fully vaccinated and boosted individuals consistently show lower rates of cases, hospitalizations, and deaths compared to unvaccinated groups, validating the protective effect of vaccination.
- **Temporal Trends:** Time-series analysis highlighted key periods of spikes and declines, often aligning with known variant waves or public health policy changes.
- **Data Quality:** The dataset demonstrated high consistency and completeness, with only minimal missing values and no significant gaps in weekly reporting.

- **Distribution Patterns:** Outcome rates show significant variability among unvaccinated populations, with broader distributions and higher outliers, emphasizing greater vulnerability and unpredictability.
- **Normalized Insights:** By using age-adjusted and population-normalized metrics, the analysis ensured fairness in comparisons and revealed deeper patterns often obscured in raw counts.

6. Future Scope:

Future enhancements of this analysis could focus on expanding its analytical depth and interactivity. Incorporating predictive modelling techniques, such as time-series forecasting or classification models, would allow for estimating future case counts, hospitalization risks, or mortality probabilities across vaccination groups. This could support health agencies in preparing for potential surges.

7. References:

- [1] Microsoft Excel Documentation: <https://support.microsoft.com/en-us/excel>
- [2] Python Pandas Documentation: <https://pandas.pydata.org/docs/>
- [3] Matplotlib Documentation: <https://matplotlib.org/stable/contents.html>
- [4] Seaborn Documentation: <https://seaborn.pydata.org/>